

ABHISHEK SUGAM

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SUMMARY

- Product design engineer with expertise in electromechanical assemblies, product conceptualization & rapid prototyping
- Demonstrated expertise in injection molding, die casting, CNC machining, stamping, upset forging, and sheet metal design
- Collaborated with multifunctional teams like Audio, Optics, RF to convert system requirements into product tenants

SKILLS

- CAD: Proficient in PTC Creo with Windchill PLM, NX, Solidworks
- Analysis: ANSYS Workbench & MATLAB.
- Technologies: Audio Subassemblies, Touch displays, IoT Sensor systems, Handheld devices.

PROFESSIONAL EXPERIENCE

Product Design Engineer | Amazon

June 2021- Present

- PD lead responsible for system architecture, product definition from pre-concept phase to mass production for next generation Edge AI capable Alexa+ devices
- Spearheaded the design and implementation of advanced thermal management systems, successfully meeting stringent thermal ID requirements and setting new benchmarks for an Alexa device
- System architect and DRI for Audio subsystem, PCBs and RF components for Echo Dot & Echo Show 15 & 21
- Conceptualized and implemented a new cost-down audio subsystem design that resulted in over \$25M in savings during the product lifecycle while keeping the device performance on par with the previous generation

Mechanical Engineer | Siemens

May 2019- May2021

- Responsible for mechanical design and layout studies for overall system architecture configuration selection for IoT Devices comprising of PIR sensors, thermal cameras & environmental sensors.
- Designed the next-gen IoT device which was three times smaller, has two times smaller installation time and 30% cheaper
- Identified a gap in the market and created a new IoT device, which wound up generating tens of Millions of \$ in revenue

Mechanical Engineer | Microsoft

Feb 2018- May 2019

- Responsible for mechanical design of electronic enclosures & touch display modules for the 50" & 86" Surface Hub 2
- Responsible for characterizing optically clear resin (OCRs) & process development for optically bonding large touch displays.
- Responsible for mura characterization and mitigation using Radiant vision cameras & JND metrics for edge & corner Muras

Mechanical Systems Engineer | Seattle Monorail

May 2017- Feb 2018

- Designed a new Charge Collector and touch screen-based controls for the Classic Seattle Monorail trains.
- Strategically transform complex business requirements into robust system specifications, driving innovation and ensuring all products adhere rigorously to FTA regulatory standards.

Mechanical Design Intern | Ubi Interactive Inc

Jan 2017- April 2017

- Designed and prototyped a device that impacts touch capabilities to any screen using IR emitters and cameras
- Responsible for the design and analysis of Mechanical and electrical components & enclosures for the Emitter and Camera.
- Owned and executed all the environmental tests like vibration & shock and thermal analysis of electronic enclosures

Senior Student Partner | Microsoft

Nov 2012- April 2018

- Student Developer Advocate for Microsoft, organized workshops and hackathons on Data Science, IoT & Machine Learning
- Selected among the 50 best performing student partners in the world for three consecutive years 2016-2018
- Created a Microsoft developer community from ground up at UW through blog posts, social media, meetups, and workshops

ACADEMIC EXPERIENCE:

Powertrain & Battery pack Engineer | UW EcoCAR3

Oct 2015 – Dec 2017

- Led a team of 25 students to design and manufacture a hybrid powertrain and the battery pack for 2016 Chevrolet Camaro
- Manufacturing the high tolerance transmission components using CNC mills, lathes and heat treatment processes and fabricated the battery Pack using sheet metal processes, injection molding and vacuum forming.

Graduate Research Assistant, Cell Biomechanics Lab | UW

Nov 2015 – June 2016

- Responsible for the design of a portable electrical chamber & stretchable membrane to study the stem cell conversions
- Performed tolerance analysis to meet dimensional build objectives and produced drawings as per ASME Y14.5 standards.

Team lead, Chassis Team | Formula SAE 2015

Oct 2014 – May 2015

- Led the engineering team with the goal of developing the chassis and suspension system of a Formula SAE car
- Designed and optimized the Chassis based on multibody dynamic simulations for various loading conditions

Team lead, Roll Cage & Transmission Team | Baja SAE 2014

Mar 2013 – Feb 2014

- Led a group of 18 engineers developing the roll cage and transmission system for the all-terrain vehicle
- Designed and fabricated the roll cage and created drawings and BOMs for the transmission systems Assembly

EDUCATION

University of Washington, Seattle, USA

Graduated 2017

Master of Science in Mechanical Engineering, 3.61/4

Birla Institute of Technology, Mesra, India

Graduated 2015

Bachelor of Engineering in Mechanical Engineering, *First Class with Distinction*